

Dawson Ren

☎ +1 (314) 882-1693 | ✉ dawsonren@gmail.com | 🔗 linkedin.com/in/dawson-ren/ | 🌐 dawsonren.github.io/

Education

Northwestern University, McCormick School of Engineering, Evanston, IL 2024 – Present
PhD Student, Industrial Engineering and Management Science
Advisor: Daniel Apley
TA: Statistical Learning for Data Analysis, Machine Learning II (Master's), Probability
Northwestern University, McCormick School of Engineering, Evanston, IL 2020 – 2024
Bachelor of Science with Honors in IEMS (Cum Laude)

Research Experience

Local Variable Importance Measures (VIMs) August 2025 – Present
Graduate Research Assistant | Mentor: Daniel Apley

- Designing explainable AI (XAI) method to interpret model predictions, enabling faster debugging and stakeholder trust in black-box machine learning models
- Benchmarking new method against existing explainable AI (XAI) techniques (e.g., LIME, SHAP) while demonstrating their shortcomings

Characterizing Grain Boundary Resistance using Machine Learning September 2025 – Present
Predictive Science and Engineering Design Fellow | Mentors: Daniel Apley and Oluwaseyi Balogun

- Extracting insights from grain boundary defect data in thermoelectric materials (Si, SnTe) to guide the manufacturing process of next-generation materials, with applications in heavy industry and space
- Building scalable data pipelines on HPC infrastructure to iteratively retrain models using active learning, reducing experimental cost for compute-intensive simulations

Sequential Decision Making for E-Commerce and Vehicle Routing August 2023 – August 2025
Undergraduate / Graduate Research Assistant | Mentors: Seyed Iravani, Sean Sinclair

- Formulated an online algorithm that learns a company's optimal free shipping threshold, poster presentation at the 2025 Midwestern Optimization and Statistical Learning Conference
- Developed heuristics for equitable vehicle routing and allocation in nonprofit operations

Professional Experience

Caterpillar June 2022 – August 2023
Machine Learning Intern, Applications Intern

- Engineered machine learning model to interpret oil sample severity levels for heavy machinery
- Improved classification accuracy by 15% over existing rules-based system, saving thousands of analyst hours at 70+ fluid sample labs around the world
- Presented use case for generative AI to the 20+ members of the Analytics and AI Center of Excellence, leveraging chain-of-thought prompting to summarize vehicle inspections

Skills

Python, Scientific Computing (scipy, scikit-learn, pandas), Deep Learning (PyTorch), SQL (PostgreSQL, SQLite), Frontend Development (Angular, React), Cloud Computing (AWS EC2, S3, Lambda), MLOps (Weights and Biases), Deployment/Virtualization (Docker), Version Control (Git), REST APIs (FastAPI)

Leadership and Awards

IEMS Outstanding Leadership Award June 2024
Semi-finalist, Fletcher Outstanding Research Prize June 2024
Senior Residential Assistant, Northwestern University September 2022 – June 2024

- Led event planning and conflict resolution for 50 residents, supervised 14 Residential Assistants